Kyle Wright

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Summary

Mathematical physicist returning to research, aimed at the mathematical foundations of AI alignment. PhD and postdoctoral work on symmetry and duality structures in string theory, quantum and conformal field theory — generalized geometry, Courant and Lie algebroids, gauged sigma models — including two sole-authored papers in the *Journal of Geometry and Physics*. Industry experience in applied quantitative modelling under real-world constraints, most recently supporting New Zealand's electrification. Applied gauge invariance to neural network compression in a volunteer research collaboration, working directly with parameter-space degeneracy in large language models.

Research Experience

Iliad Intensive — Technical AI Alignment Sep – Oct 2026
London Initiative for Safe AI (LISA) London, UK

- Completed a four-week full-time taught programme in foundational AI alignment research (learning theory, interpretability, agent foundations, computational mechanics), with a particular interest in the computational mechanics of world models — belief-state geometry and minimal predictive representations

AI Research Collaborator Oct 2025 – Feb 2026
Bradbury Group (volunteer research lab) Remote

- Investigated symmetry-based compression of large language models: identified the gauge invariance structure of GPT-2's attention weights, derived gauge-invariant variables in an affine formulation, and implemented a Fisher-weighted SVD compression scheme in PyTorch that outperformed standard SVD at matched parameter budgets at the head level, where the gauge symmetry acts

Postdoctoral Researcher Aug 2019 – Aug 2021
New York University Abu Dhabi Abu Dhabi, UAE

- Applied group-theoretic and algebraic methods to classify symmetry structures in quantum field models, developing analytical frameworks to characterise their representations and physical constraints

Postdoctoral Researcher Aug 2018 – Aug 2019
Australian National University Canberra, Australia

- Developed algebraic and geometric frameworks for symmetry and duality structures in string-theoretic sigma models; sole-authored publication unifying Lie algebroid gauging conditions via generalized geometry

Publications

- K. Wright, *Lie Algebroid Gauging of Non-linear Sigma Models*, J. Geom. Phys. **146** (2019) 103490 — sole author
- K. Wright, *Generalised Contact Geometry as Reduced Generalised Complex Geometry*, J. Geom. Phys. **130** (2018) 331–348 — sole author
- P. Bouwknegt, M. Bugden, C. Klimčík, K. Wright, *Hidden Isometry of "T-duality without Isometry"*, JHEP **08** (2017) 116
- K. Wright, A. S. Bradley, *Stochastic Longevity of a Dark Soliton in a Finite-Temperature Bose–Einstein Condensate*, arXiv:1104.2691 (2011)

Professional Experience

Grid Investment Advisor Aug 2022 – Nov 2024
Transpower (New Zealand's Electricity Grid Operator) Wellington, New Zealand

- Led the cost–benefit analysis underpinning a NZD $83m Major Capex Proposal to expand the national transmission grid for New Zealand's electrification, securing regulator approval
- Built Python-based probabilistic models — stochastic renewal-process simulation, forecasting, and economic optimisation under uncertainty — to quantify long-term investment needs for a decarbonising, electrifying grid, with documented assumptions and reproducible pipelines

Independent Research — Data Science & Machine Learning Aug 2021 – Aug 2022

- Built an event-driven backtesting framework in Python; developed latent-state and time-series models (HMMs, ARIMA, GARCH) with out-of-sample validation across regime changes

Education

PhD, Theoretical Physics 2013 – 2018
Australian National University Canberra, Australia

- Thesis: *Generalised Geometries and Lie Algebroid Gauging in String Theory* — generalized contact and coKähler structures via reduction, and global aspects of non-isometric Lie algebroid gauging; produced sole-authored publications in the *Journal of Geometry and Physics* and a collaboration in *JHEP*

BSc (Hons 1st class), Physics & Mathematics 2005 – 2010
University of Otago Dunedin, New Zealand

- A+ honours theses on stochastic dynamics of dark solitons in finite-temperature Bose–Einstein condensates and on mathematical foundations of quantum theory

Teaching

Lectured an honours (4th-year) course in conformal field theory, Australian National University.
Tutored 3 undergraduate courses across 9 offerings, Australian National University.

Skills

Mathematics: Differential and generalized geometry; Lie theory, algebroids, and representation theory; quantum and conformal field theory, renormalization group; path integral and perturbative methods; stochastic processes and probabilistic modelling

Programming & ML: Python (PyTorch, NumPy, SciPy, pandas, statsmodels, scikit-learn), MATLAB, Git; reproducible research pipelines